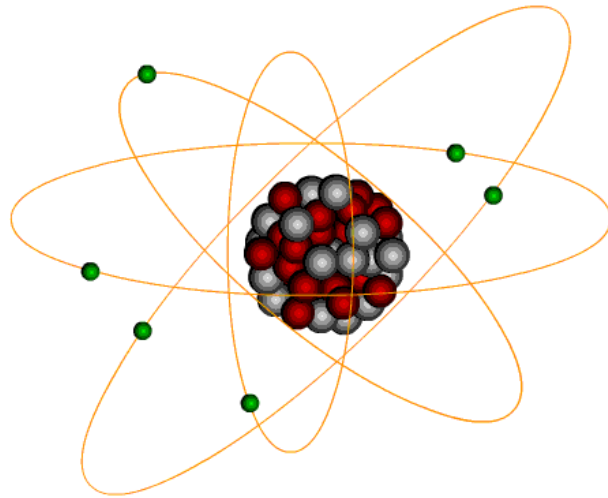


11.6 Inquires about the nuclear energy and its uses



Sri Lanka Atomic Energy Board

Contents

- **Nuclear and Chemical Reactions**
- **Binding Energy of Nucleus**
- **Binding Energy per Nucleon and Stability of Nucleus**
- **Nuclear Fission and Fusion**
- **Operation of Nuclear Power Plant**
- **Nuclear Fuels – Fissile Materials**
- **Physical Effects of Nuclear Weapons**
- **Nuclear Weapons**



Nuclear and Chemical Reactions

- Atoms are attached with each other with Chemical Bonds
- But Nucleons in a Nucleus attached with strong nuclear bonds.
- Therefore very strong Energy is released by a Nuclear reaction than a Chemical reaction

Chemical Reaction



Nuclear Reaction



Binding Energy of a Nucleus

- Nucleus of the Atom construct with **Neutrons** and **Protons** can be identified as **Nucleons**
- The energy required to separate all the **Nucleons** (Neutrons and Protons) in a Nucleus from each other is known as the “**Binding Energy of a Nucleus**”



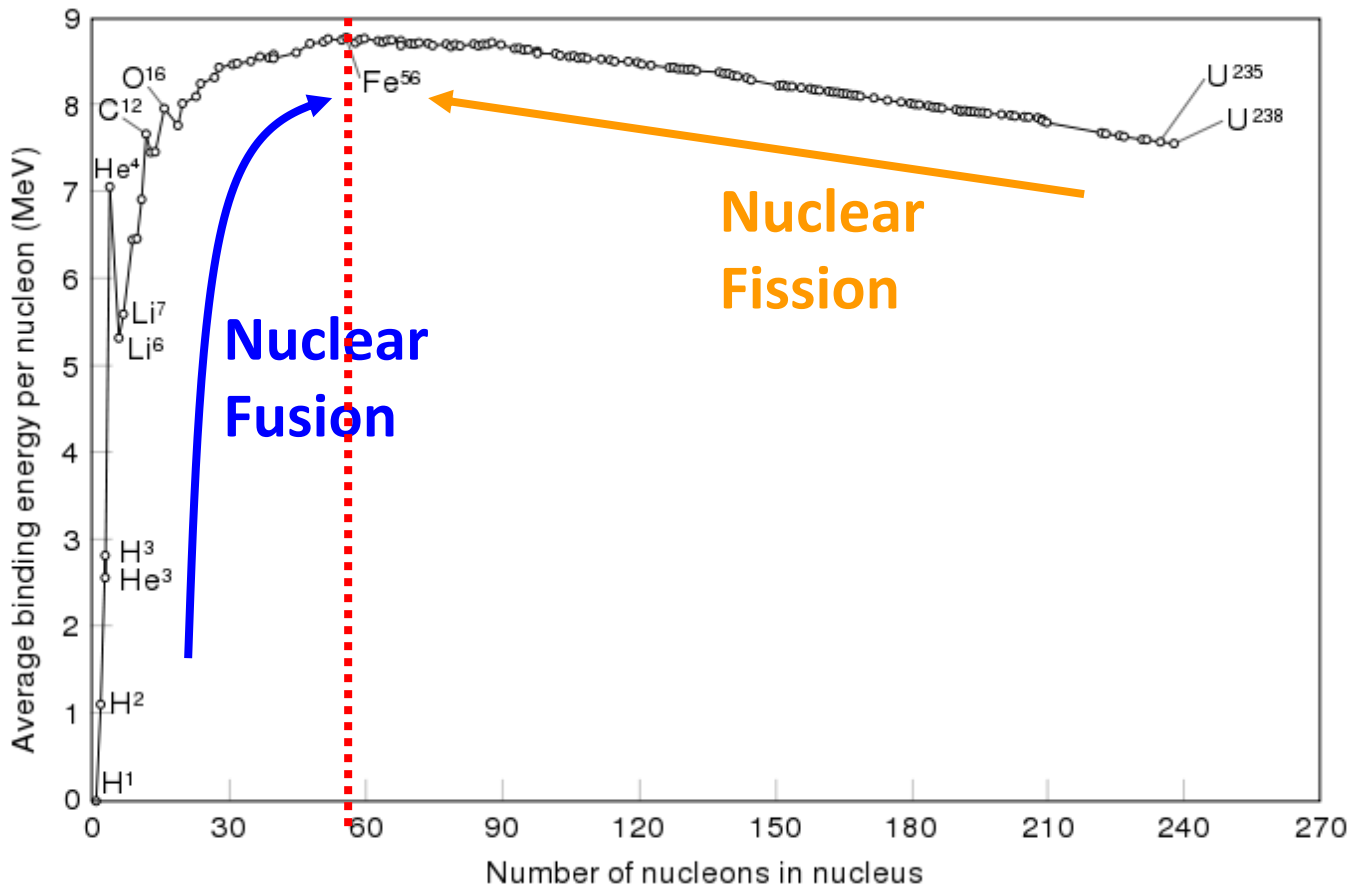
Binding Energy per
Nucleon

=

Binding Energy of the Nucleus
Number of Nucleons



Binding Energy per Nucleon



- Nuclear Fission or Fusion emerge to produce the Nucleus which has higher Binding Energy per Nucleon

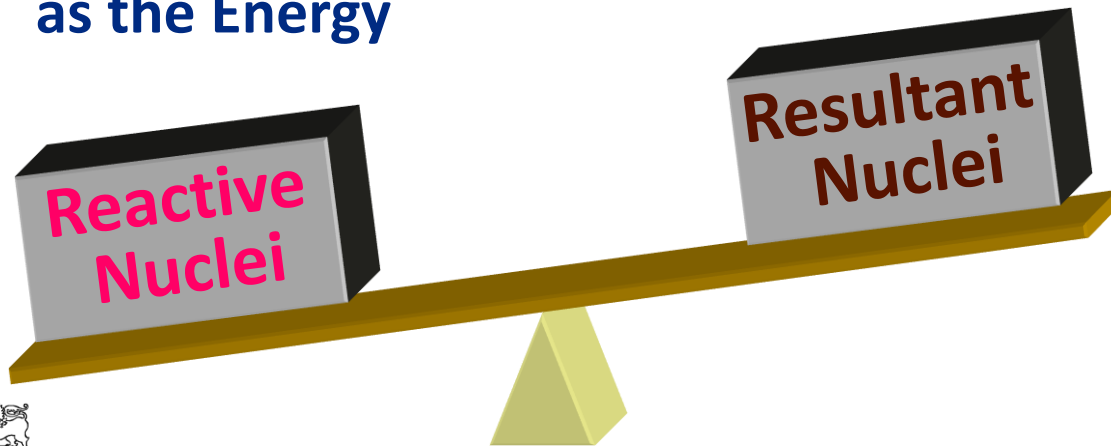


Nuclear Reactions

❖ Nuclear Fusion

❖ Nuclear Fission

- Mass of **Resultant Nuclei** of a Nuclear Reaction is less than the mass of **Reactive Nuclei**
- Einstein's Equation describe ($E = mc^2$) and Lost Mass release as the Energy



E = Energy

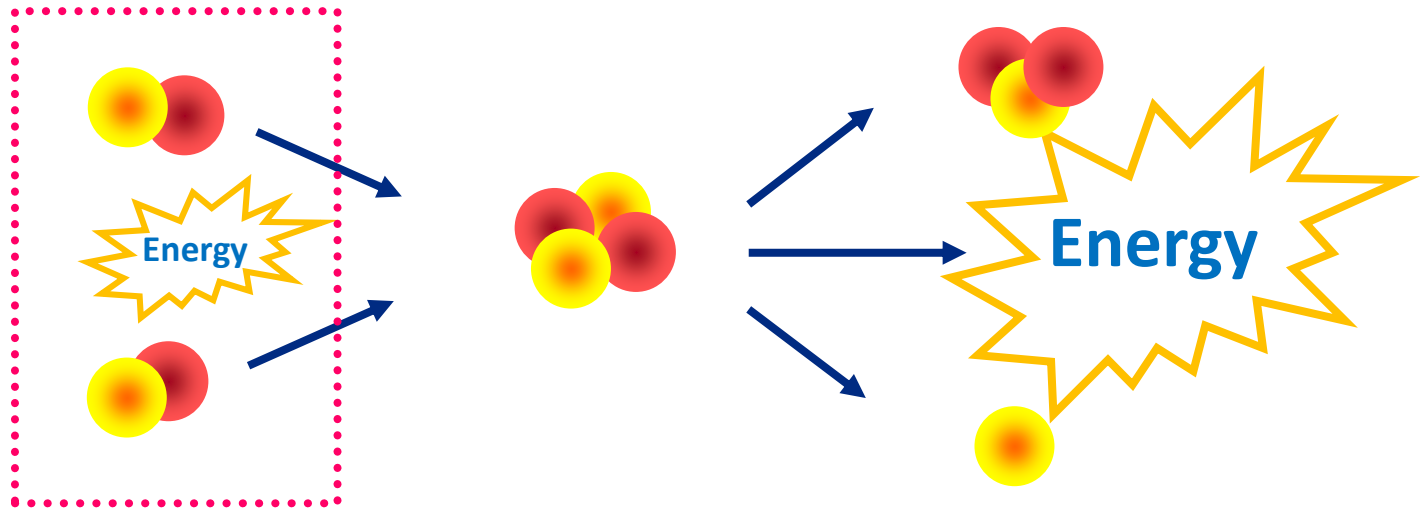
m = Lost Mass

C = Velocity of Light
 $\sim 3 \times 10^8$

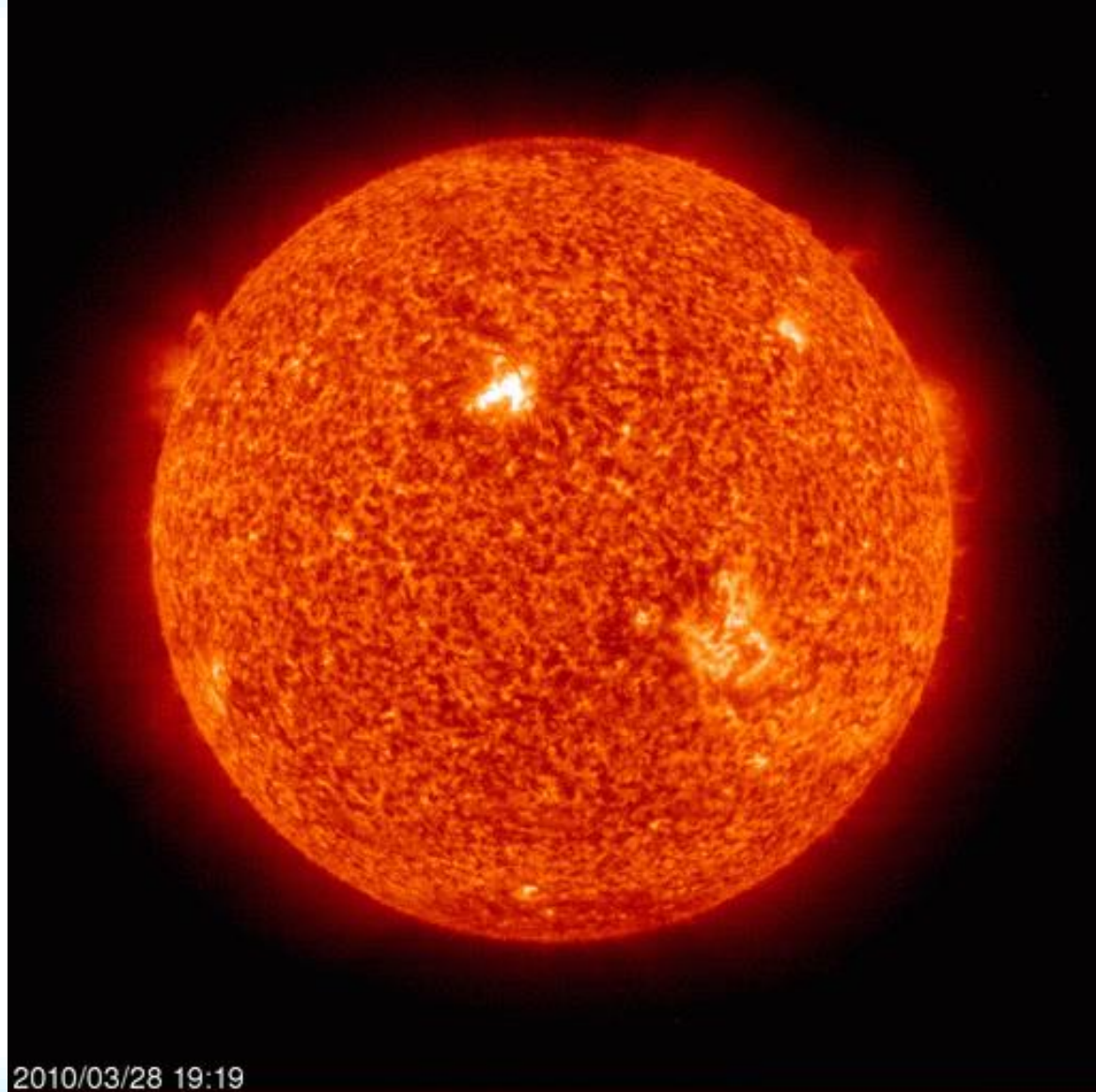


Nuclear Fusion

- Small nuclei combined together and forming a large nuclei is known as **Nuclear Fusion**
- Large thermal energy is required initially to combine small nuclei, and can gain more powerful energy as output
- Practically this task is difficult, because higher thermal energy is required initially

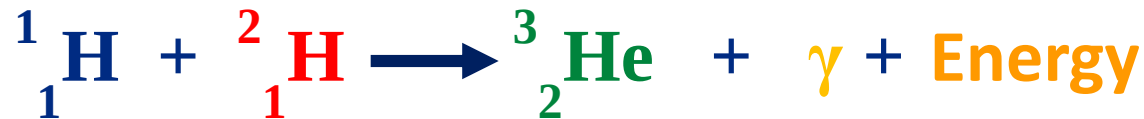
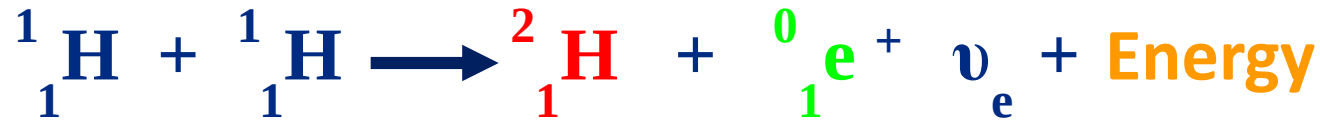


Fusion Reaction in the Sun

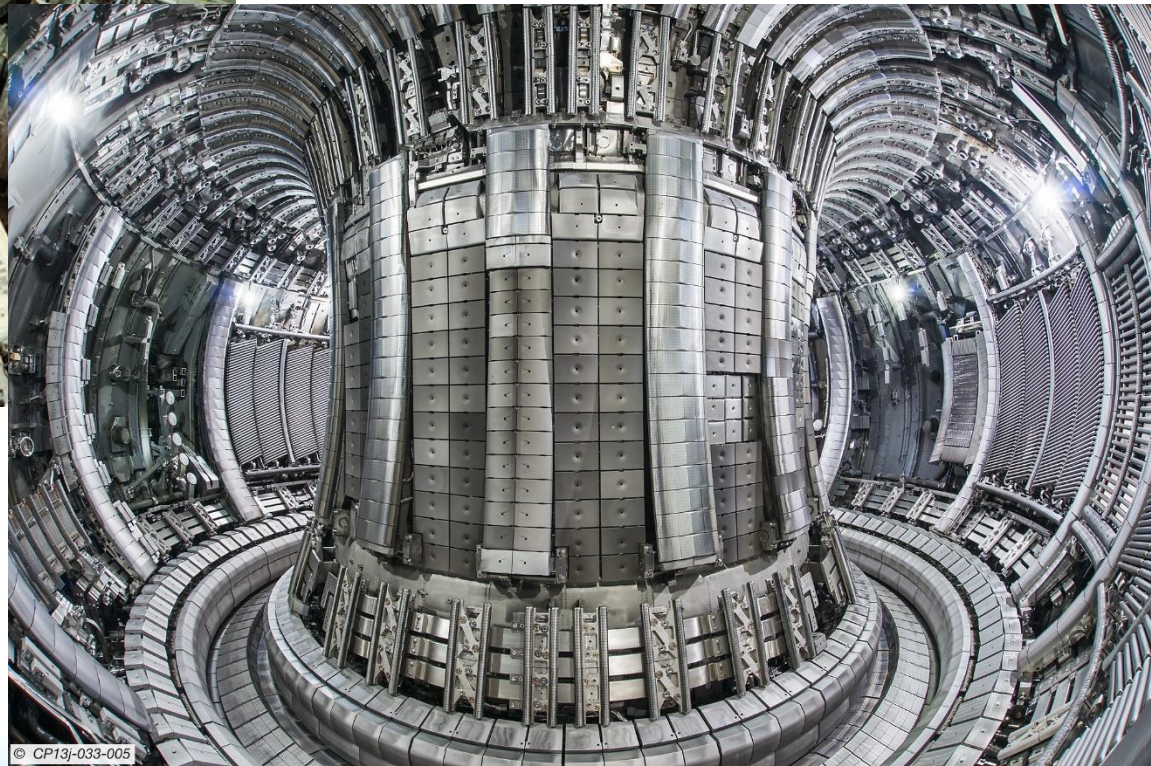
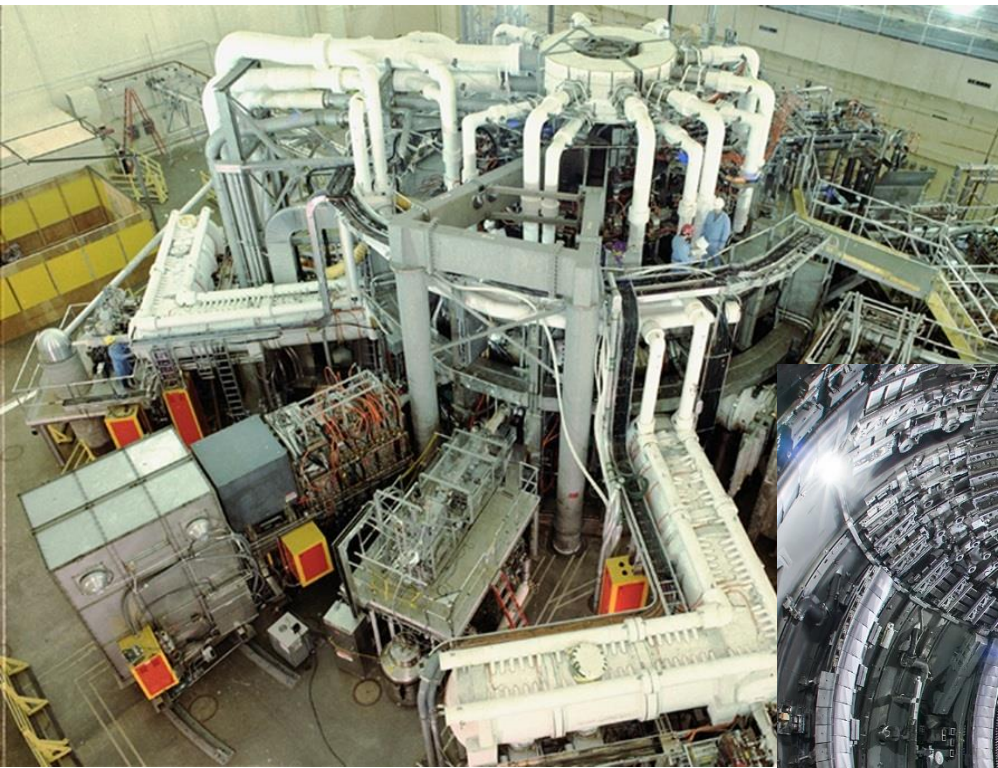


Solar Energy

- Solar Energy is produced by the Nuclear reaction of Hydrogen fusion
- Examples for Fusion

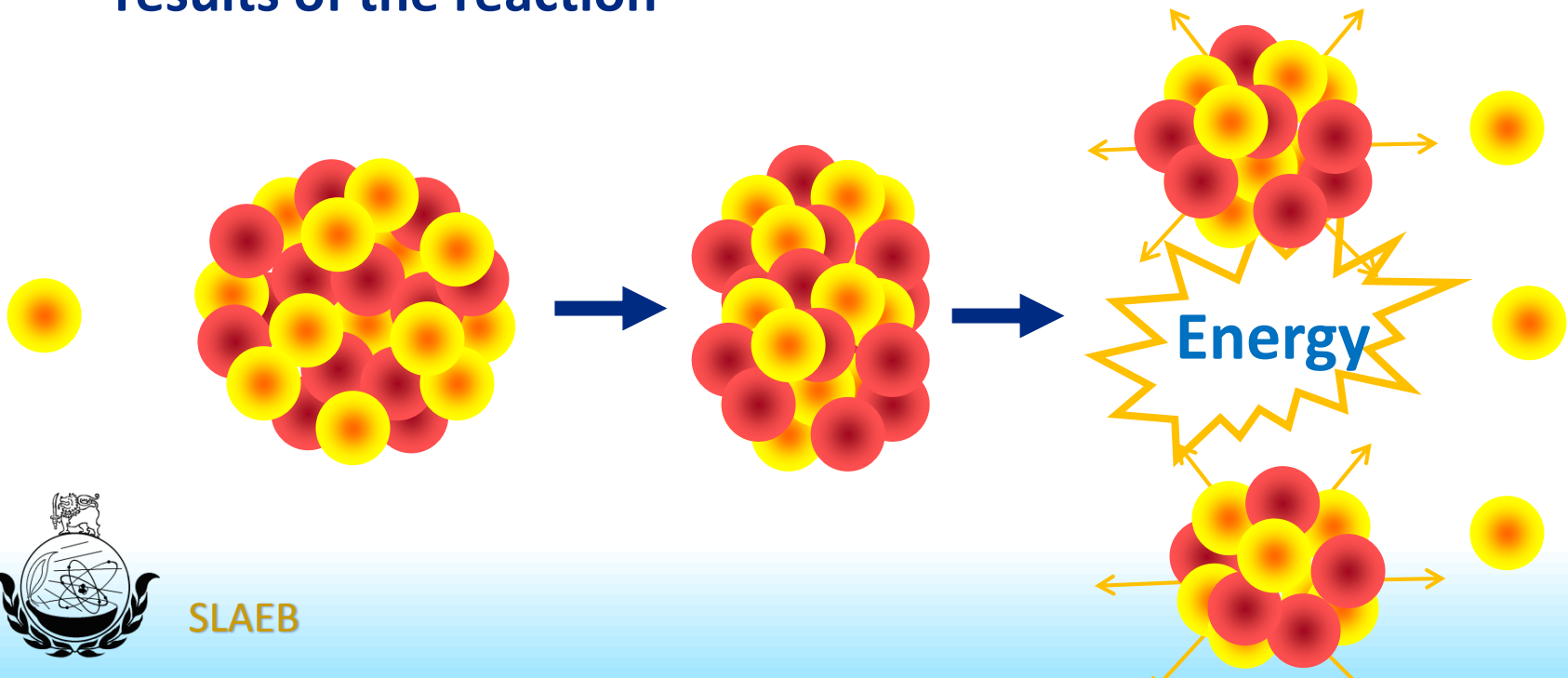


Tokamak Nuclear Fusion experimental research reactor in Tokai-mura, Japan



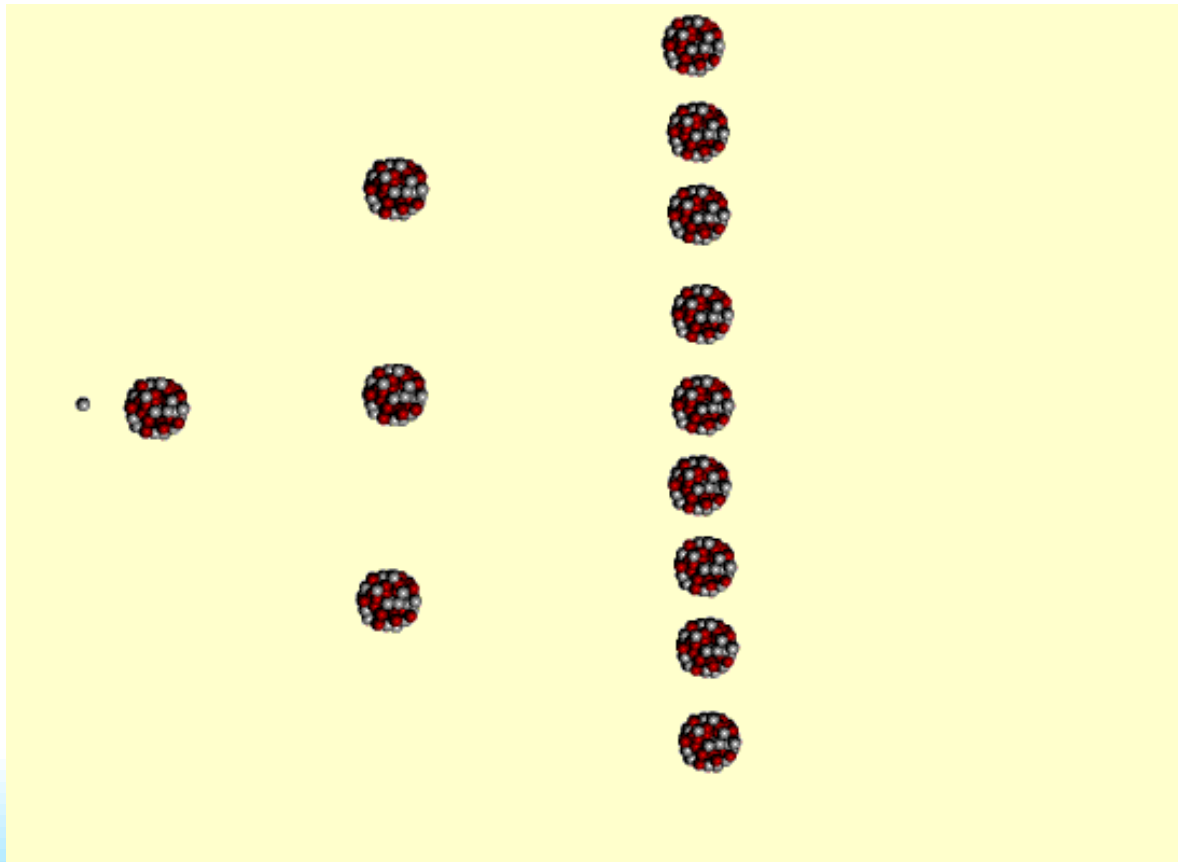
Nuclear Fission

- An unstable heavy Nucleus breaking in to small Nuclei by releasing energy is called as **Nuclear Fission** reaction
- Collide of Neutron with a unstable heavy Nuclei cause to start a Fission Reaction.
- Lost mass of the Reaction, released as Thermal Energy
- Unstable Radioactive Nuclei and Neutrons are produced as results of the reaction



Fission Chain Reaction

- Resultant Neutrons cause to initiate chain reactions and produce huge thermal energy and unstable radioactive isotopes like ^3H , ^{131}I , ^{129}I , ^{137}Cs , ^{90}Sr



If these chain reactions are not controlled

- Huge thermal energy with harmful radioactive materials are released to the environment instantly.
- This is the basic process of a Nuclear weapon

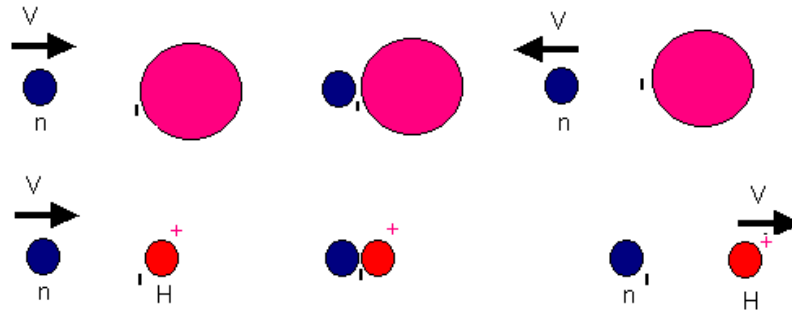
Nuclear Explosion



Control the Chain Reaction

- Chain reaction can be controlled by absorbing additional Neutrons produced with the **Fission Reaction**.
- Water with Hydrogen Atoms (Protons) can be used to moderate fast neutrons
- Graphite rods can be used to absorb the moderated neutrons

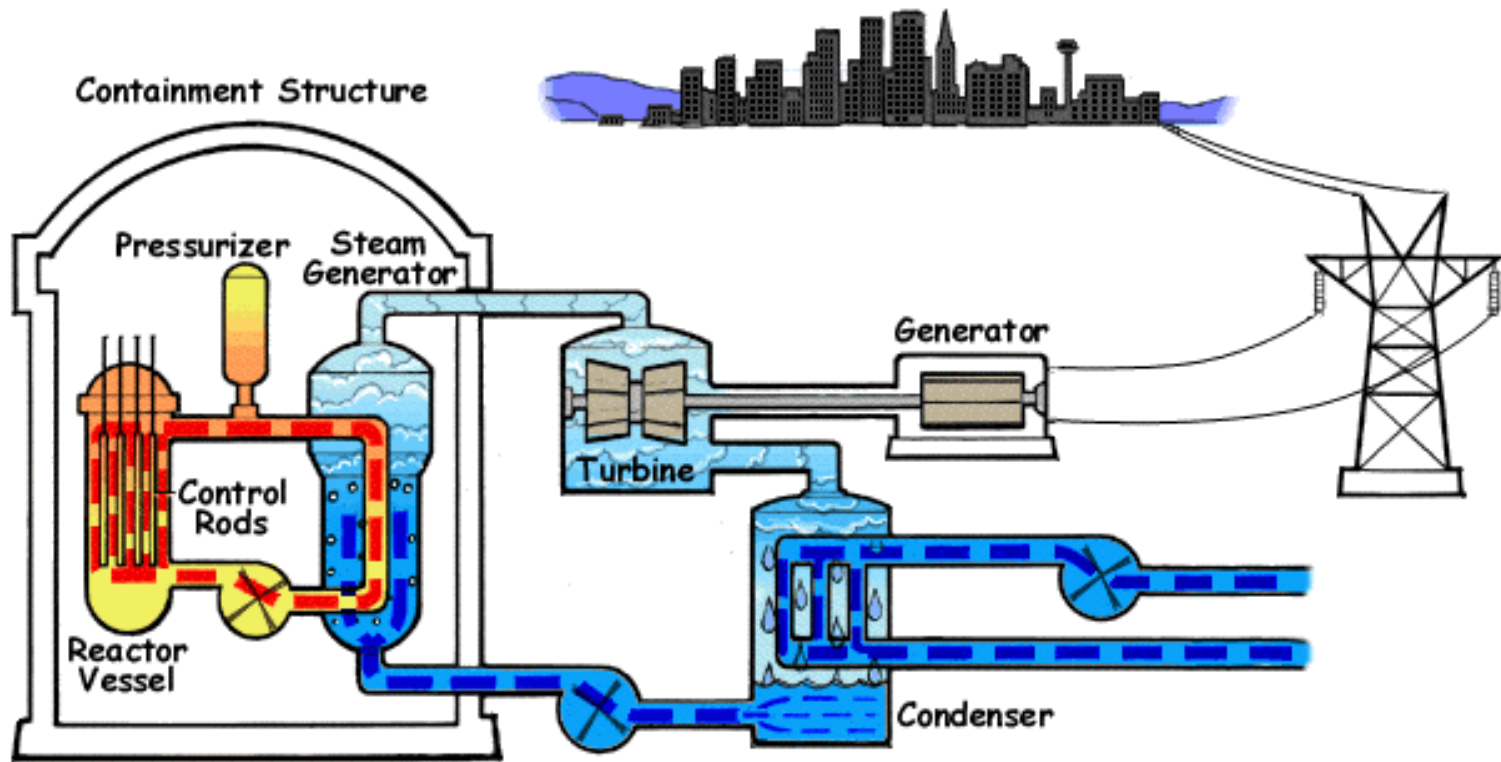
Moderate fast Neutrons



Neutron Capture



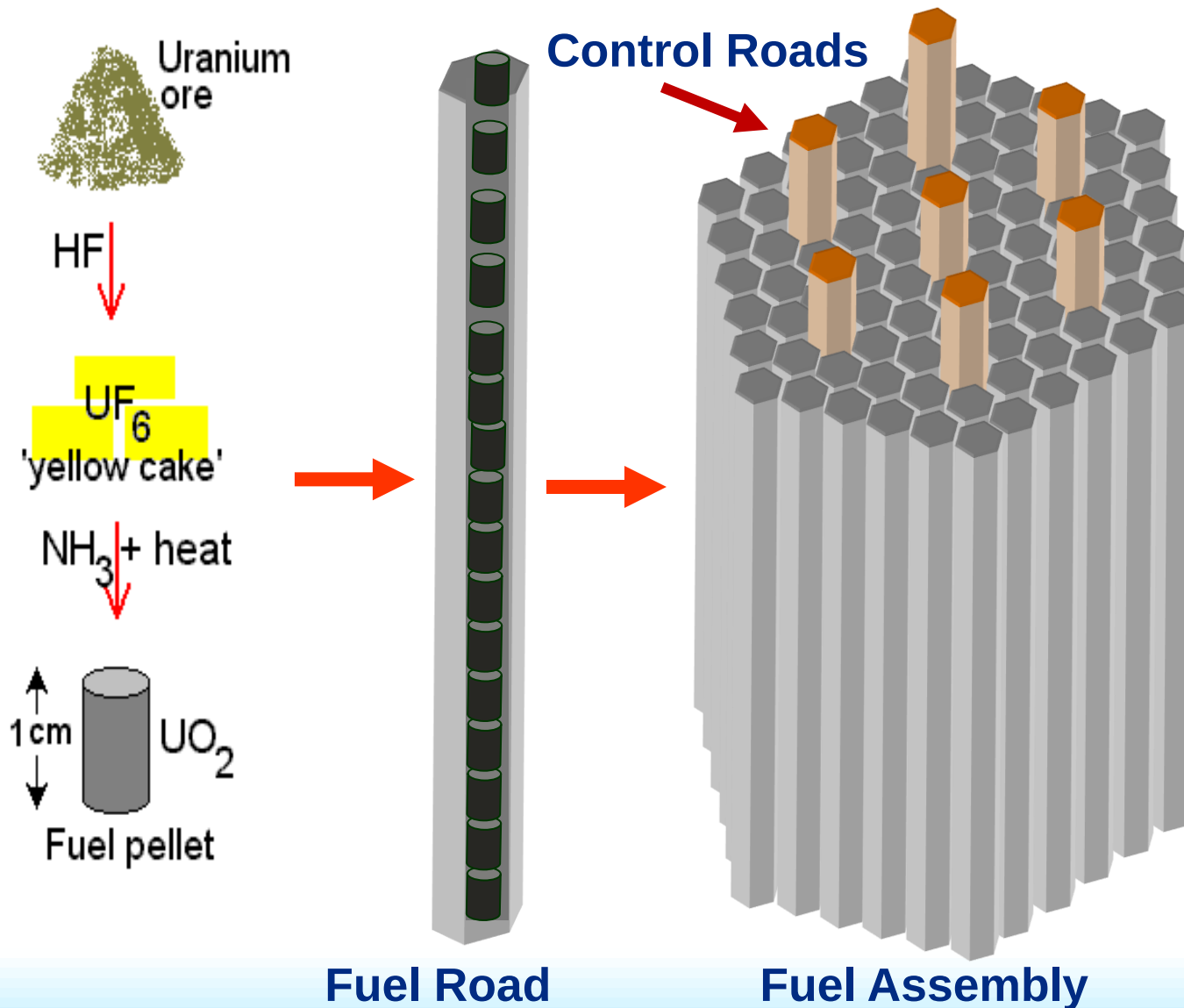
Nuclear Power Generation



- Steam is generated by the Energy produced with the **Nuclear Fission** reactions in Reactor Vessel
- Generated steam is used to produce Electricity by rotating the Turbine and the Generator



Reactor Core



Reactor Core

Water

1. As a Coolant
2. As a Moderator

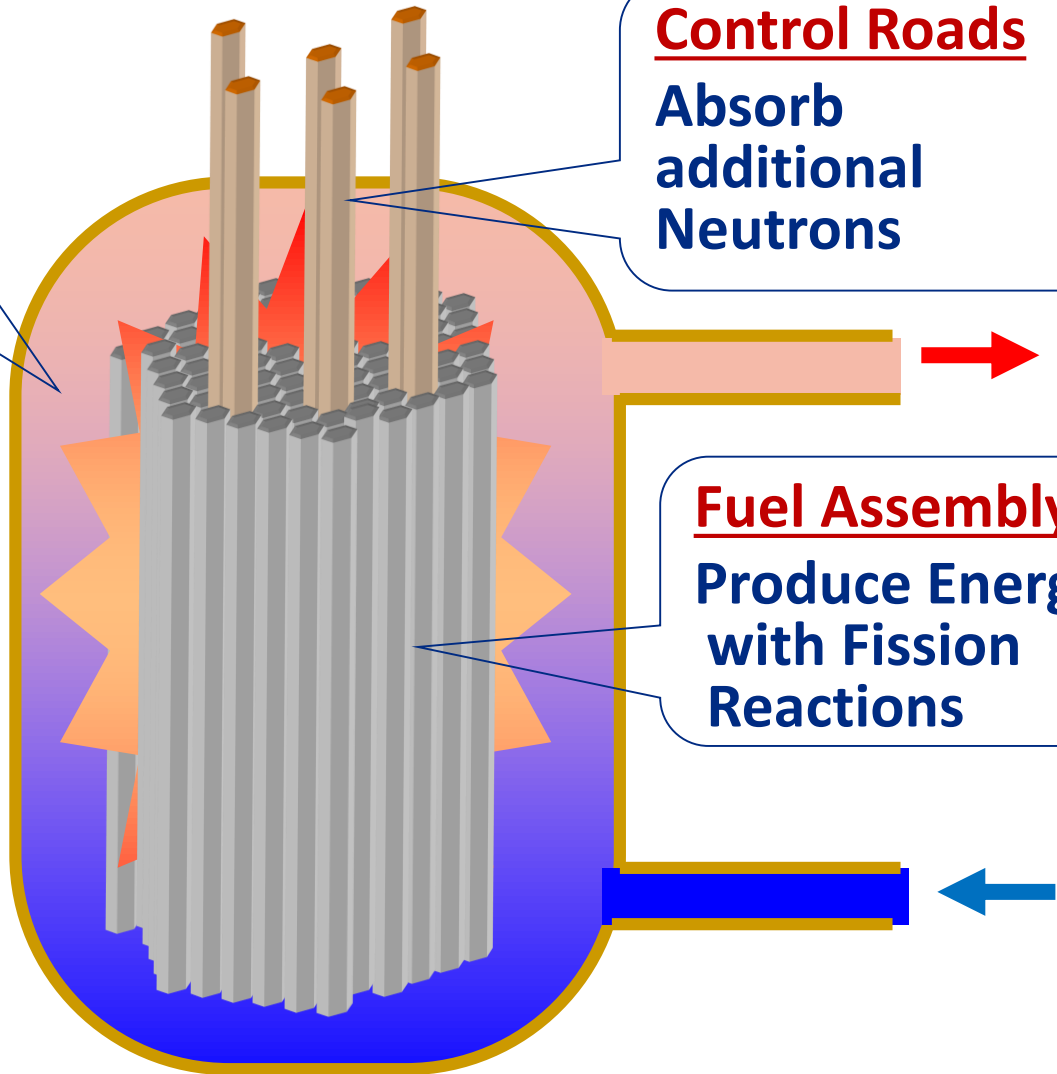
Energy Produced inside the Reactor Vessel can be increased or decreased by moving the control rods up and down

Control Rods

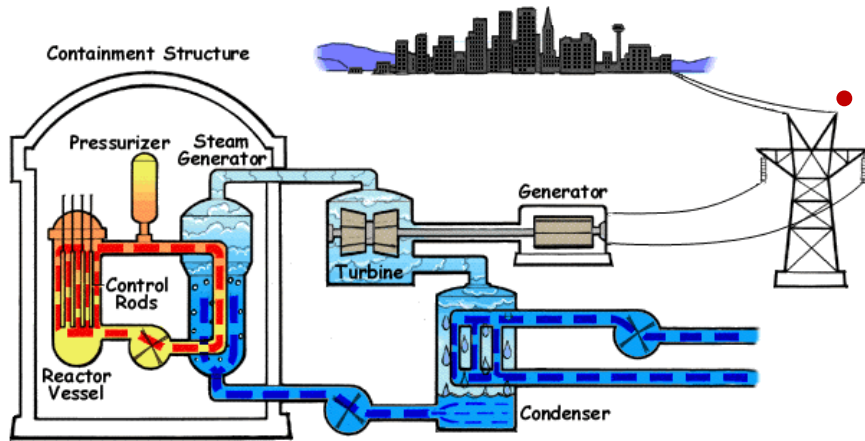
Absorb additional Neutrons

Fuel Assembly

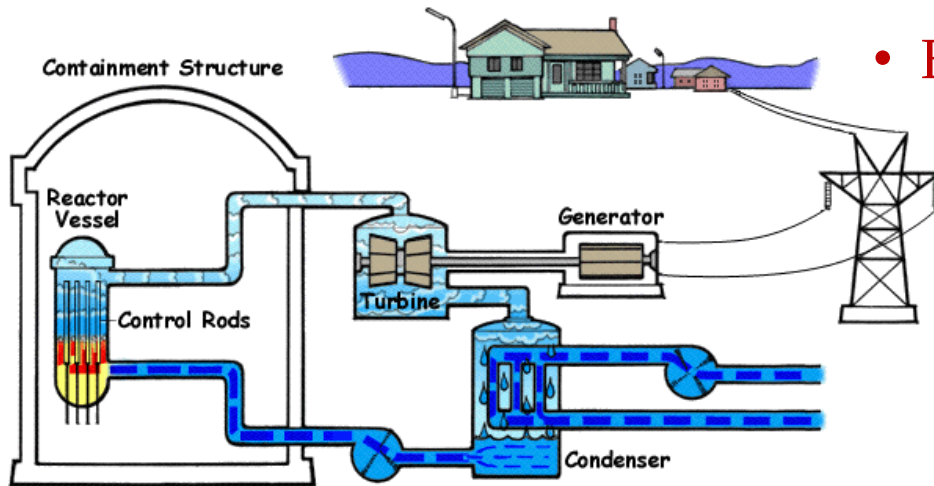
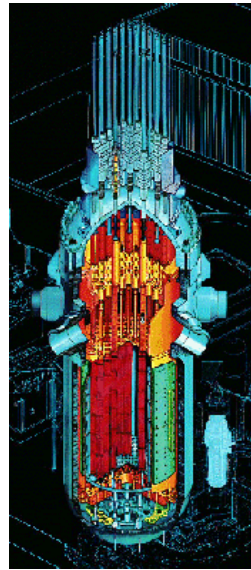
Produce Energy with Fission Reactions



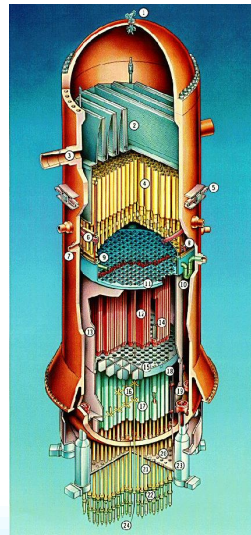
Nuclear Power Plant Types



- PWR –
Pressurized Water Reactor



- BWR –
Boiling Water Reactor



Nuclear Power Plant

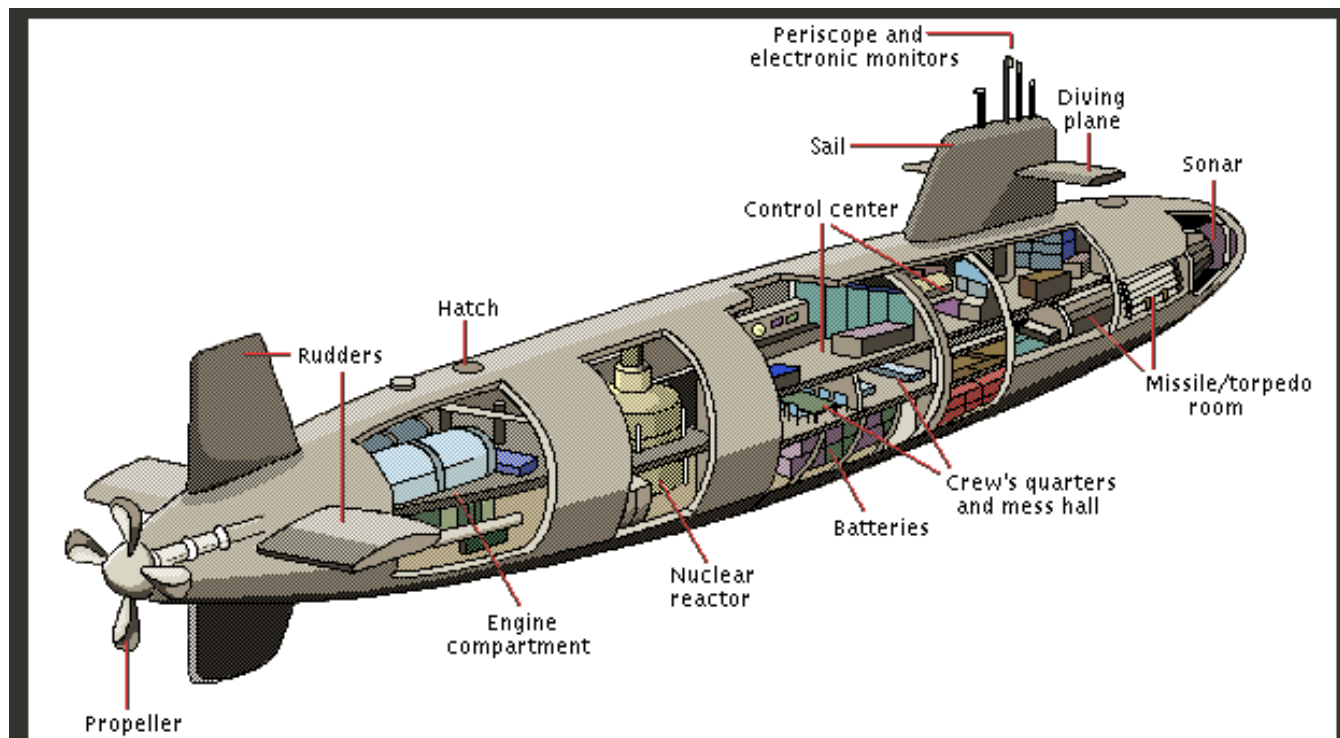
Nuclear Power Plants

The Palo Verde Nuclear Power Facility in Arizona, like other nuclear power plants, was built to harness nuclear energy for controlled use by humans. The first full-scale nuclear power plants began operating in 1956 in England and in 1957 in the United States. Subsequently, several other countries followed suit. Nuclear power is a controversial energy source: it is inexpensive and creates no air pollution, but the radioactivity released during accidents at nuclear power plants has caused deaths and environmental damage.

Jeffrey Muir
Hamilton/Gamma Liaison



Nuclear Submarine



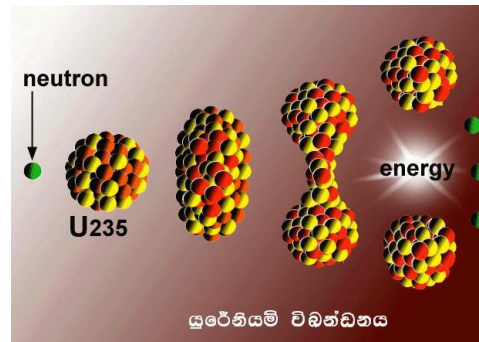
Nuclear Submarine

Nuclear submarines consume a relatively small amount of energy and make very little noise. Because they carry their energy source with them, nuclear submarines are able to travel at least 640,000 km (400,000 mi) without refueling. The nuclear reactor provides energy in the form of heat, which is converted to electricity by the generators in the engine compartment. A propeller is used to send the submarine through the water, whereas rudders (horizontal rudders are also called diving planes) guide the submarine through maneuvers. The periscope and other monitors mounted on the sail give the crew information about the surface while the submarine stays safely beneath. A modern submarine is capable of carrying several missiles, torpedos, or nuclear warheads that may be fired from beneath the water to strike targets sometimes thousands of miles away (launching tubes not shown here).



Nuclear Fuel

- All the large Nuclei cannot fission with Neutrons



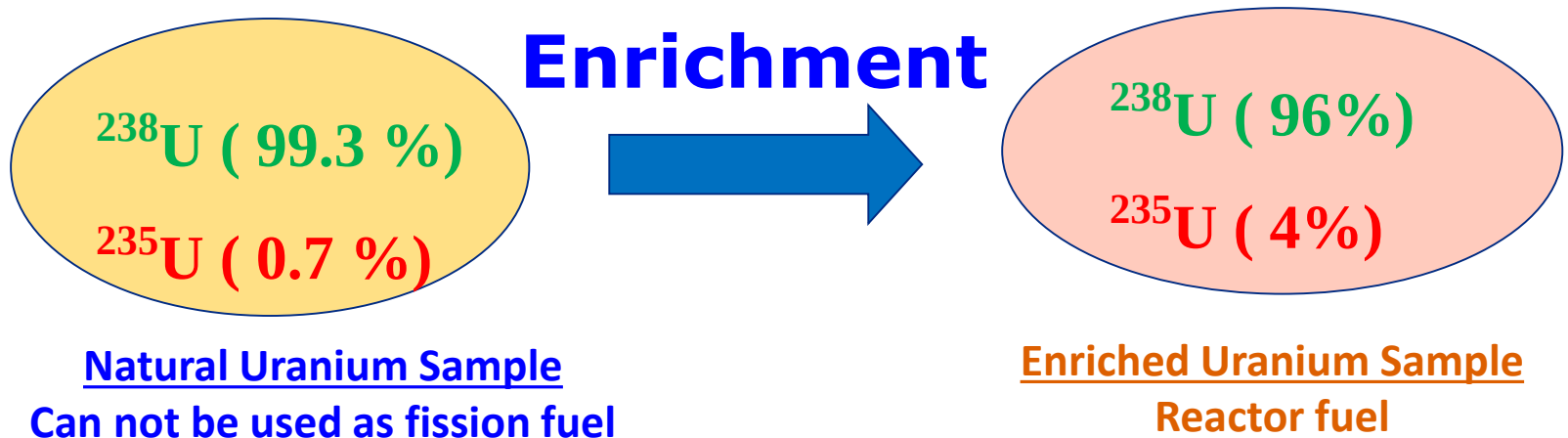
- The only few Nucleus that can fission with Neutrons called as **Fission Fuel**
- Fission fuels can be divided into two categories
 - **Natural Fission Fuel** - **U-235**
 - **Artificial Fission Fuel** – **Pu-239 & U-233**



Natural Fission Fuel

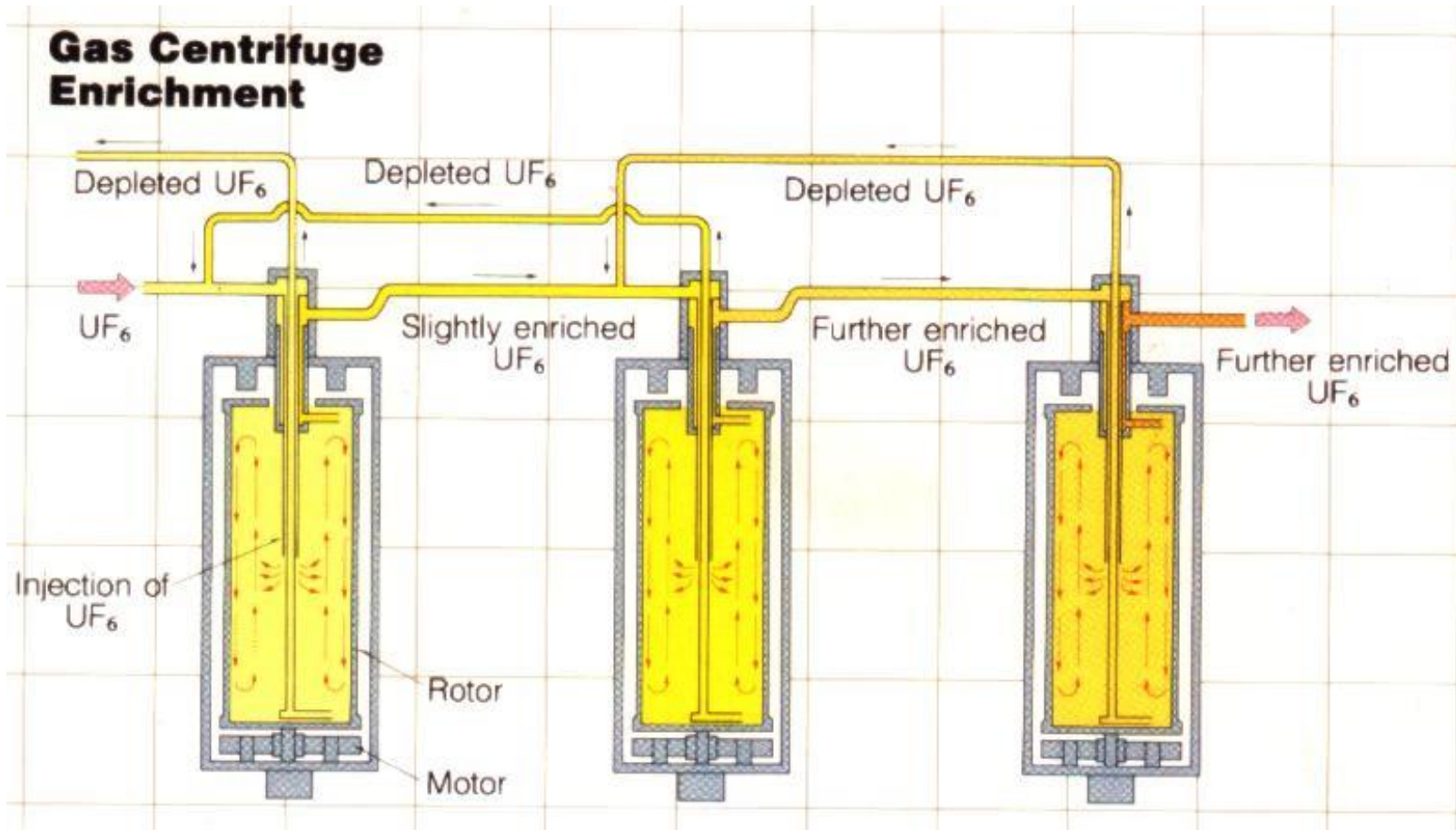


- Uranium-235 is the Natural Fission fuel
- But the percentage of U-235 in a Natural Uranium sample is 0.7%
- 3% - 4% Enrichment is needed to use it as a Fission Fuel in Nuclear Power Plant





Enrichment Process

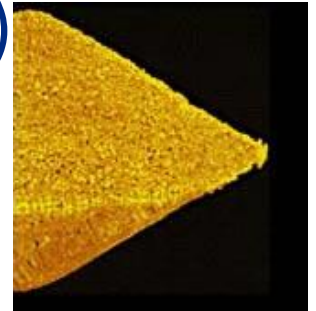


Different types of Enrichment methods are used to enrich ^{235}U and Gas Centrifuge is one method.



Uranium Processing

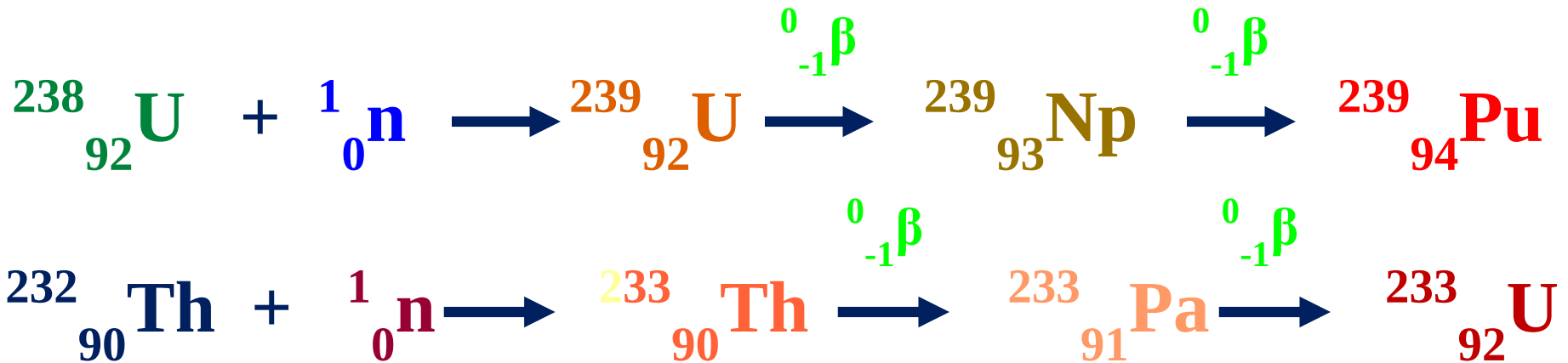
- Uranium is mined as ore from open pits or deep shaft mines, often with the help of extracting solutions
- At nearby mills, ore is crushed and U is extracted, leaving behind radioactive tailings
- Extracted U is then leached (with sulfuric acid) forming a concentrate known as “yellowcake” (aka Uranium oxide U_3O_8)
- Yellowcake is then turned into UF_6 gas, which can be cooled to a solid for easier transport



Artificial Fission Fuels



- Plutonium- 239 and Uranium-233 are Artificial Fission fuels
- These fuels are generated from the natural Uranium -238 and Thorium- 232 as follows



Why Nuclear Bomb so much worse than a TNT bomb?

- Amount of heat and light energy released is 1000 times greater
- Explosion is accompanied by invisible, penetrating and harmful radiation
- After explosion, radioactive fallout remains and continues to damage living things for days → weeks → years



Ground level view of
Hiroshima cloud



Nuclear Weapons are Scary too!

- Most of the lasting effects are due to radiation, so are odorless and colorless
- Genetic damage and cancers can take 20 or more years to develop
- A single bomb can kill 100,000 people and destroy an entire city
- It does not take much nuclear material to create a big explosion
- However, it does take considerable engineering to make a bomb that works



Physical Effects of Nuclear Weapons

- **Thermal**

- Fireball → Firestorms
- Mushroom Cloud

- **Initial (prompt) Radiation**

- Alpha particles (${}^4\text{He}^{++}$)
- Beta particles (e^+ and e^-)
- Gamma-rays (γ)
- Neutrons (n)

- **Pressure Blast Wave**

- Buildings collapse

- **Fallout**

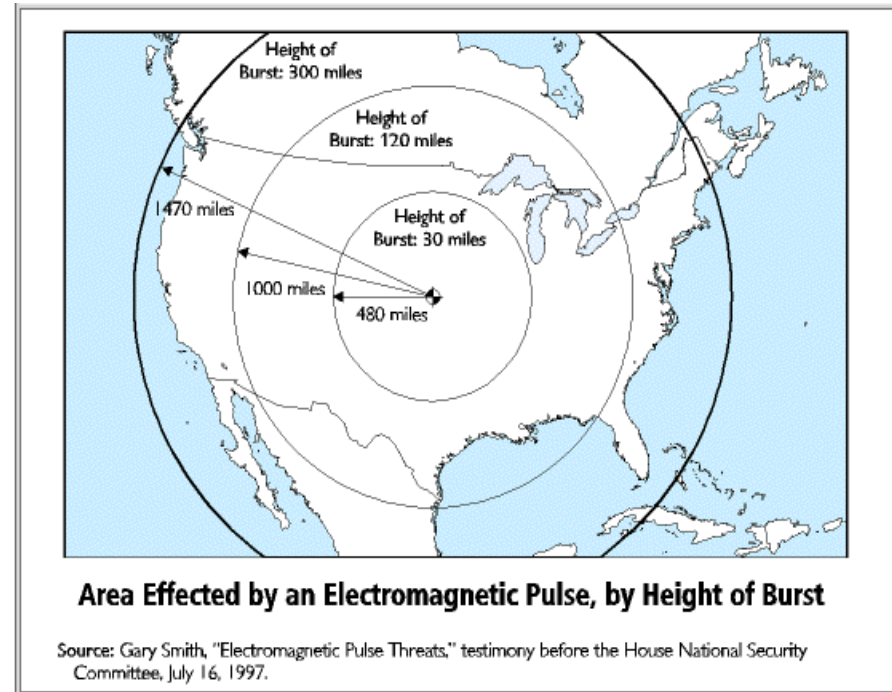
- Radioactive fragments which stick to air particles or dirt that is sucked up mushroom stem
- 80% falls back down in first day
- 90% falls back down in first week
- 10% lasts weeks → years



Physical Effects of Nuclear Weapons

- **Electromagnetic Pulse**

- Strongest for very high bursts
- Gamma-rays ionize air → electrons
- Electrons create large currents in air
- Currents are picked up by power lines
- Power surges shut down grid, destroy attached electrical devices



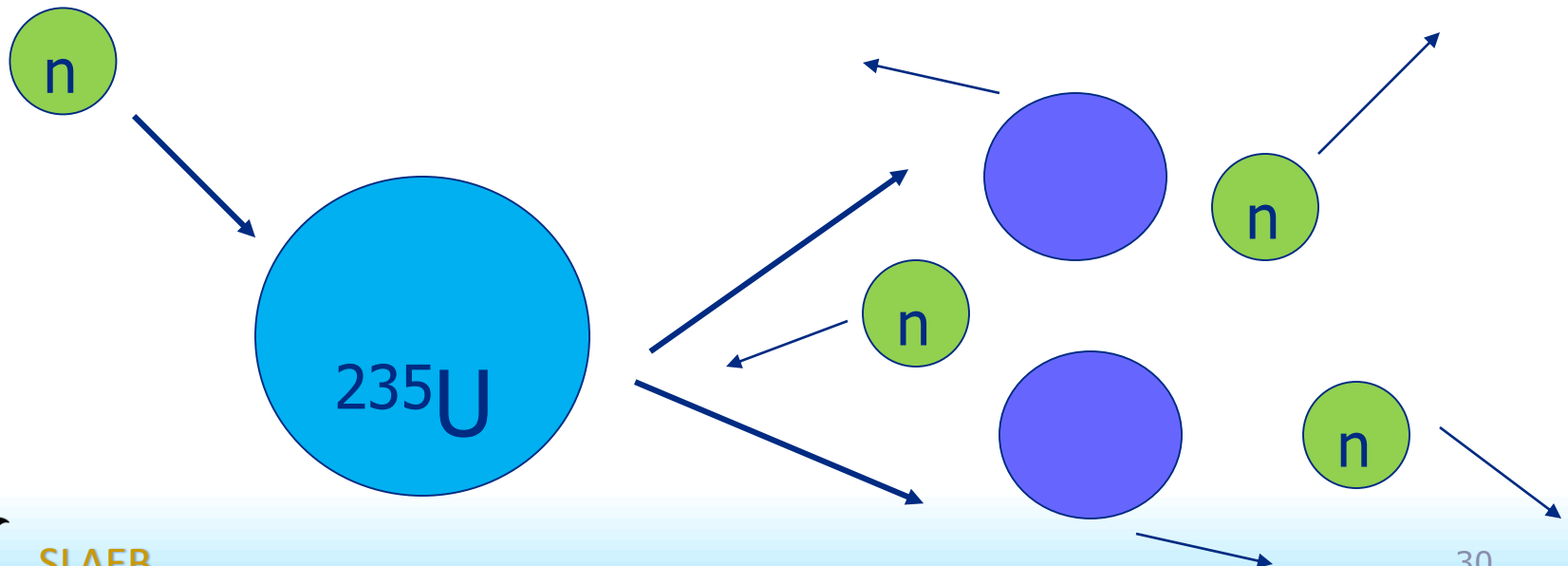
1.4 Mton airburst in 1962
knocked out lights in Hawaii over
1000 miles away



Fission Weapons

- **Fission**

- Releases energy in elements heavier than Iron
- Bombard **U** or **Pu** with neutrons, they split into fragments, releasing energy
- “A” bombs

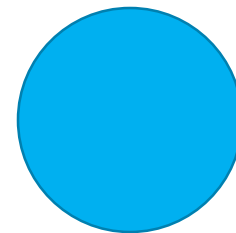
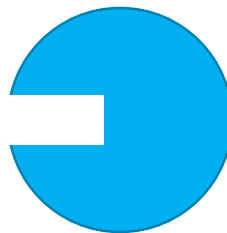


Critical Mass → Chain Reactions

- When a large enough mass of either ^{235}U or ^{239}Pu is assembled, a self-sustaining chain reaction results after the first fission is produced.
- The minimum (“critical”) mass of fissile material that can sustain a nuclear chain reaction depends on the density, shape, and type of fissile material, as well as the effectiveness of any surrounding material (called a reflector or tamper) at reflecting neutrons back into the fissioning mass.



Subcritical



Supercritical



The first “A” bombs

- **Trinity – Gadget (16/07/1945)**
 - Alamagordo test range in New Mexico
 - 20 kTon yield
- **Little Boy (06/08/1945)**
 - Hiroshima
 - 15 kTon yield
- **Fat Man (09/08/1945)**
 - Nagasaki
 - 20 kTon yield



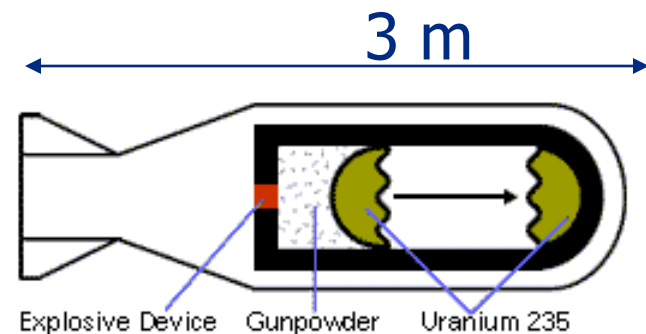
How to make an “A” bomb

- Use $>90\%$ ^{235}U
- Squeeze and confine evenly
- Reflect neutrons back into ^{235}U
- Use initial explosive device to trigger



A-bomb dome

Little Boy (Hiroshima 06/08/1945)



“Fat Man” Style of A-bomb

- High explosives are arranged to form an imploding shock wave which compresses the fissile material to super criticality.

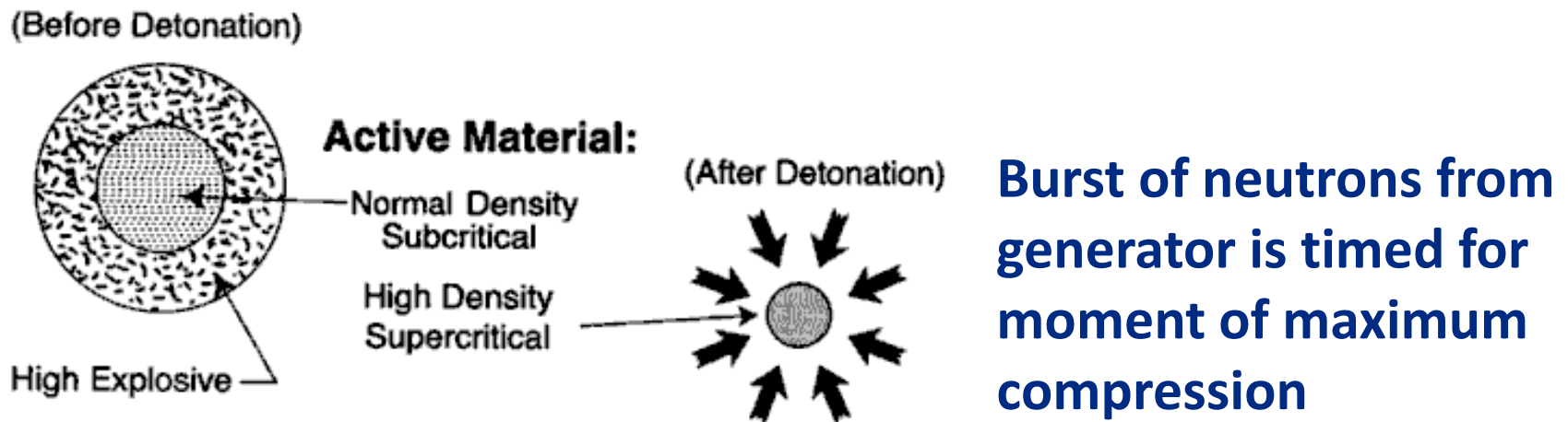
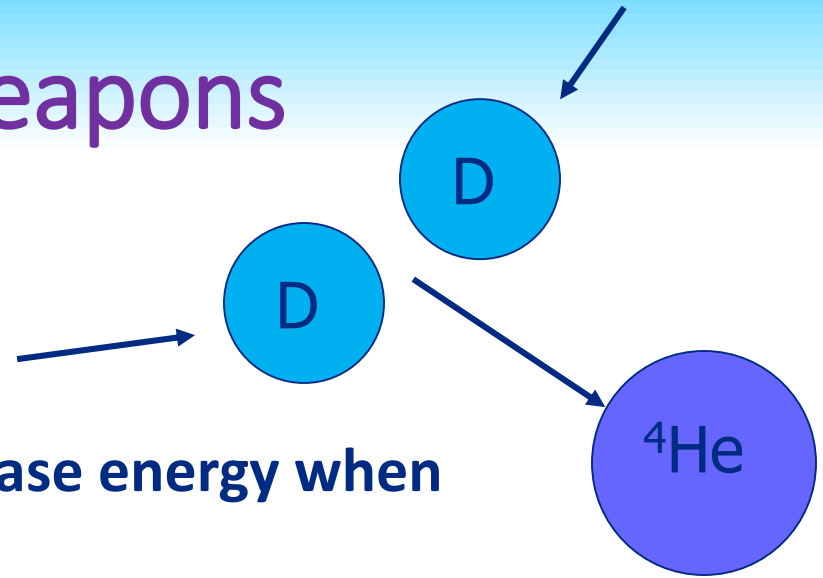


Figure 2-VIII. Implosion Assembly Principle

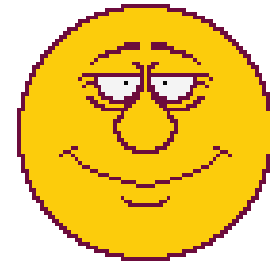


Fusion Weapons



- **Fusion**

- Elements lighter than Iron release energy when combined
- Deuterium, Tritium, Lithium
- Reactions that occur inside Sun
- “H” bombs



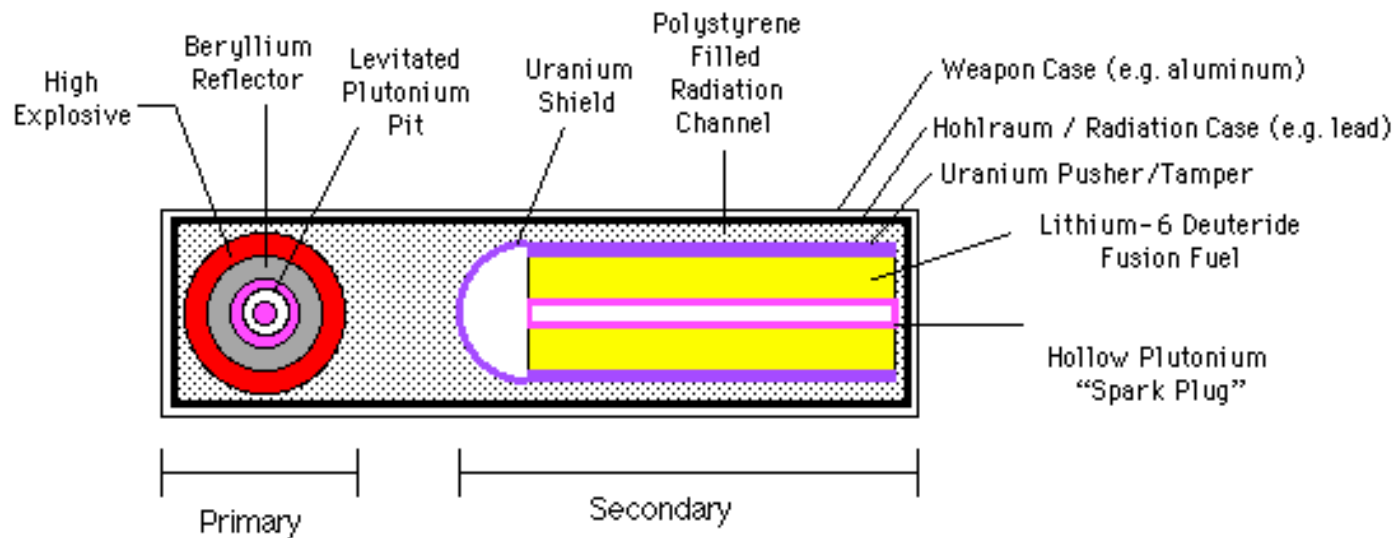
- **Thermonuclear Reactions**

- Heat from reaction increases reaction rate, so less fuel is needed → “efficient” bomb



The “secret” of the H-Bomb

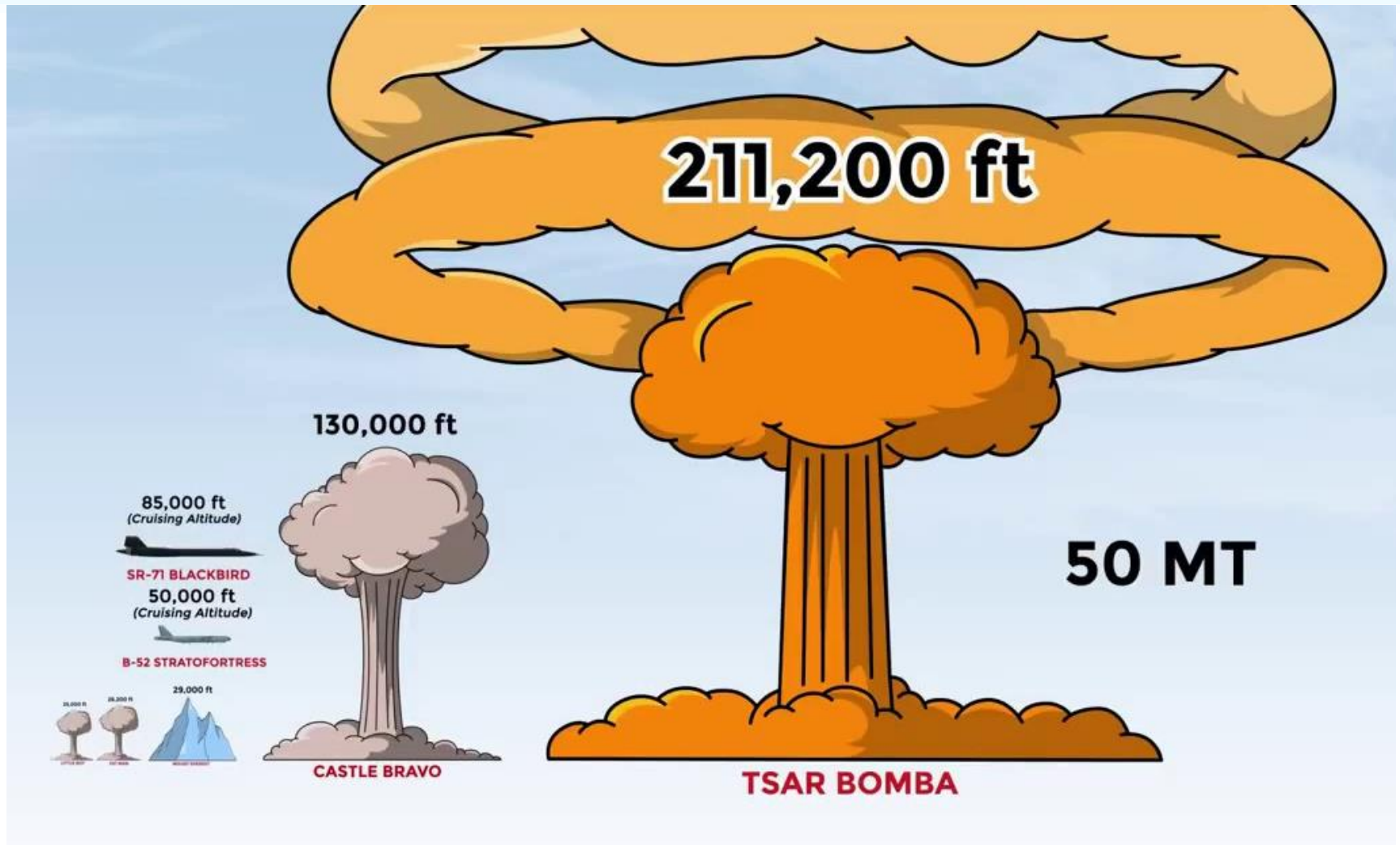
- At the high temperatures of a fission bomb 80% or more of the energy exists as soft X-rays
- The X-rays heat a polystyrene channel, causing plasma which can compress and ignite the second (fusion) stage before the expanding (1000 km/sec) primary disrupts it.



How Big are the Weapons?

- 1 kTon = 1000 tons = 2,000,000 pounds of TNT equivalent
- ~2 pounds of ^{235}U $\rightarrow$ 20 kTons
- Today's warhead is 100-200 kTons
- Largest underground burst: 4.5Mtons
- Largest airburst: 58 Mtons
- Over 1700 known tests since 1945





Who has Nuclear Weapons?

Thermonuclear

Fission

US (2700)

Warheads

<http://www.ucsusa.org/assets/documents/nwgs/Wordwide-Nuclear-Arsenals-Fact-Sheet.pdf>

Israel(80)

Russia (4500)

UK (150)

N. Korea
(<15)

China(180)

France (300)

India(50)
Pakistan(60)



Detection of Nuclear Material

- Nuclear Materials
 - Tritium = ^3H = very heavy Hydrogen (1p + 2n), used in fusion weapons
 - Deuterium = ^2H = heavy Hydrogen (1p + 1n), used in fusion weapons
 - Uranium: ^{238}U is >99% in nature ^{235}U is ~0.7% in nature – major ingredient in fission weapons
 - Plutonium: ^{239}Pu is not found in nature, used in fission weapons
- Difficult to Detect – (Not emit Strong Gamma Energy lines)
- Expert Knowledge on Alfa, Beta and Gamma Spectroscopy is needed





Thank You

